Cloud Computing:

- Cloud computing is the delivery of services like:
  - Servers
  - Storage
  - Network
  - Databases
  - Monitoring tools
  - AI
- These services are delivered over the internet, and work on the principle "Pay-As-You-Go".
- Pay-As-You-Go, means that you will only pay for the services that you use.
- Cloud computing is defined as *renting the IT services* over the internet.
- In Cloud computing, the core services are:
  - Compute (CPU + RAM)
  - Storage
  - Network

Cloud computing VS On-premises

On-premises

- ✔ Within your own premises (house, building, campus, within organization).
- ✔ You need to take care of:
  - Spaces
  - Electricity
  - Cooling
  - Security
- ✔ The major advantage of on-prem is that the data stays in-house. No 3rd party intervention.

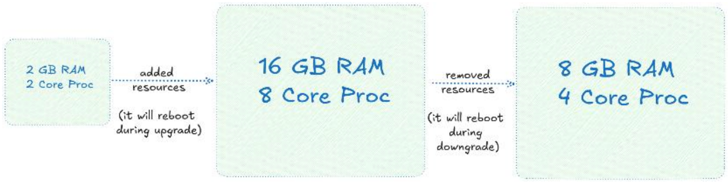
Cloud computing

- ✔ Over the internet, in some-one else's datacenter.
- ✔ Vendors like:
  - Google
  - Microsoft
  - Amazon
  - Redhat
  - VMWare
  - Rackspace
- ✔ These vendors are providing the cloud computing capabilities with a small fee.

Cloud computing features:

- On-demand
  - The required service (compute, storage, network, database, etc.) must be available immediately when it is needed.
- Scalable
  - It's the ability to increase or decrease resources.

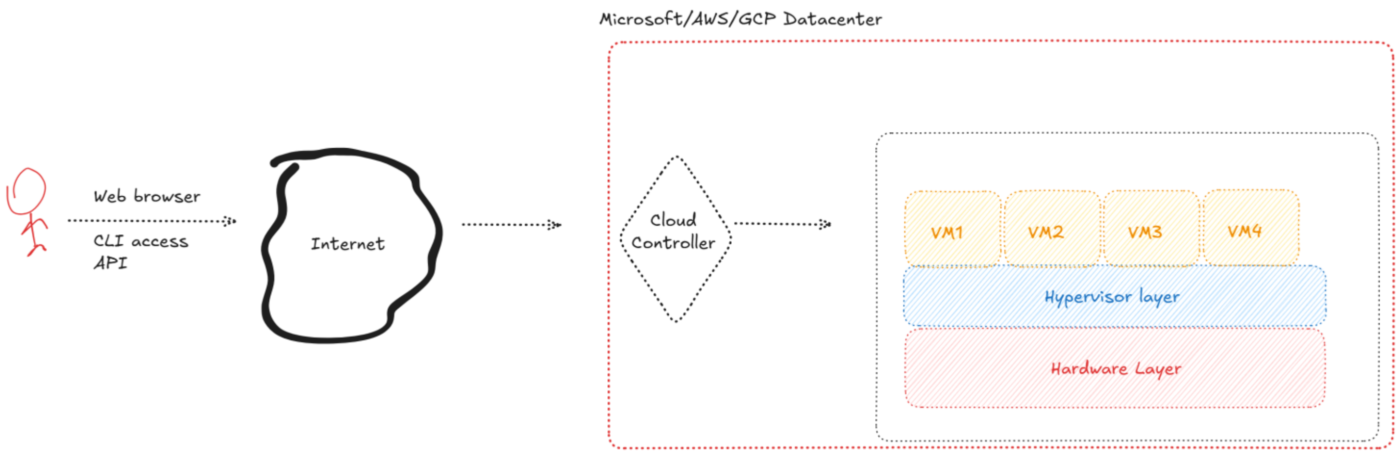
- Types of scalabilities:
    - *Vertical scaling*



        - Scaling up/down
        - Simple and faster.
        - No changes in the architecture.
        - It needs down time for upgrade and downgrade.
        - Here, we add more resources to the existing VM.
    - *Horizontal scaling*
        - Scaling out/In
        - Adding more instances to handle the traffic.
        - Scaling out → adding new instances
        - Scaling in → removing instances
        - No upper limit
        - This provides the HA (high availability) and redundancy concepts.
        - This is usually used in:
            - Amazon and Flipkart big billion sales.
            - IPL matches
                - Etc…

        - It works using load balancer to manage the traffic.
- Unlimited computation and storage
    - No limit on computation and storage.
    - No limit on storing data on cloud, what ever you upload you need to pay for that.
        - Whenever you upload a file on cloud, it makes total of 3 copies in the backend.

Cloud architecture:



Cloud deployment models:

- *Public cloud*
    - This cloud can be access by ANYONE over internet.
    - This type of cloud has multi-tenancy.
    - Example:
        - Microsoft Azure
        - Amazon web service (AWS)
        - Google Cloud Platform (GCP)
        - Redhat cloud
        - VMWare cloud
        - Digital Ocean
        - Rackspace
        - Oracle Cloud

- *Private cloud*
  - Private cloud has single-tenancy.
  - People from same organization can create their own resources under private cloud.
  - Examples:
    - Microsoft System Center Suite
    - Openstack (Linux-based) → Rackspace + NASA
- *Hybrid cloud*
  - Combination of Public and Private cloud.
- *Community cloud*
  - For specific communities
  - Example:
    - Gov cloud
      - Government cloud for govt. employees.
    - China cloud
    - Germany cloud
- *Inter-cloud models*
  - Combination of various cloud
    - AWS + Azure
    - Azure + GCP
    - AWS + GCP

Cloud Comparison Cheat Sheet

blog.bytebytego.com

aws	Azure	Google Cloud	ORACLE CLOUD
Elastic Compute Cloud (EC2)	Virtual Machine	Compute Engine	Virtual Machine
Elastic Kubernetes Service (EKS)	Azure Kubernetes Service (AKS)	Google Kubernetes Engine (GKE)	Oracle Container Engine
Lambda	Azure Functions	Cloud Functions	OCI Functions
Simple Storage Service (S3)	Blob Storage	Cloud Storage	Object Storage
Elastic Block Store	Managed Disk	Persistent Disk	Persistent Volume
Elastic File System	File Storage	File Store	File Storage
Virtual Private Cloud	Virtual Network	Virtual Private Cloud	Virtual Cloud Network
Route 53	DNS	Cloud DNS	DNS
Elastic Load Balancing	Load Balancer	Cloud Load Balancing	Load Balancer
Web Application Firewall	Web Application Firewall	Cloud Armor	Web Application Firewall
RDS	SQL Database	Cloud SQL	ATP
DynamoDB	Cosmos DB	Firebase Realtime Database	NoSQL Database
Redshift	Synapse Analytics	BigQuery	Autonomous Data Warehouse
Elastic MapReduce	HDInsight	Dataproc	Big Data
Kinesis	Streaming Analytics	Dataflow	Streaming
SageMaker	Machine Learning	Vertex AI	Data Science
Glue	Data Factory	Data Fusion	Data Integration
EventBridge	Event Grid	Eventarc	Events
Simple Queuing Service	Storage Queues	Pub/Sub	Streaming
Simple Notification Service	Service Bus	Firebase Cloud Messaging	Notifications
CloudWatch	Monitor	Cloud Monitoring	Monitoring
CloudFormation	Resource Manager	Deployment Manager	Resource Manager
IAM	Active Directory	Cloud Identity	IAM
KMS	Key Vault	Cloud KMS	Vault

AWS vs Azure

S. No	Service	Amazon Web Service	Microsoft Azure
1	Compute	Elastic Compute Cloud (EC2)	Azure Virtual Machine (VM)
2	Networking	Virtual Private Cloud (VPC)	Azure Virtual Network (vNet)
3	Storage	Simple Storage Service (S3 bucket)	Azure Storage Account
4	Users and Groups	Identity and Access Management (IAM)	Microsoft Entra ID (Azure Active Directory (AAD).

Mandatory requirements:

- Create a cloud account.
    - AWS → Phone number, Email ID, Credit or debit card (Rs 2) [RUPAY card will not be accepted].
    - Azure → Phone number, Email ID, Credit card (Rs 2).
- You need a subscription for Azure account:
    - Free trial (for 30 days, you will receive $100).
    - Pay-as-you-go account
    - MSDN subscription (for developers, freelancers, for MCT).
    - Azure student account
    - Enterprise subscription (agreement between Microsoft and Wipro).
- For AWS, only email ID (root account) and debit/credit card are required.

Ques – difference between Pricing calculator and TCO calculator.

Azure terms:

✓ Microsoft Azure pricing calculator – this gives the estimated amount for using the resources within a specific region for specific duration.
✓ Microsoft Azure TCO calculator – it's a calculator that shows the total amount that you can save after migrating to Azure cloud from on-premises. TCO = Total Cost of Ownership.
✓ Azure updates – it is a website that provides all the updates on Azure.
✓ Azure status – this website gives the updates on all Azure-based services.
✓ Azure Official documentation – Azure's official site to work with any azure resource.

To access the Azure (ARM) portal, URL: https://www.portal.azure.com

Steps for Azure:

1. Microsoft Entra ID
    ✓ How to create a user in Azure?
    ✓ How to invite a user in Azure?
    ✓ How to create users in bulk?
    ✓ How to give them access to Azure and its resources?
2. Azure Virtual Network (vNet)
    ✓ How to create a vNet using address space (192.168.0.0/16).
    ✓ How to create 2 subnets
        i. Subnet-1: 192.168.1.0/24
        ii. Subnet-2: 192.168.2.0/24
3. Azure Storage Account
    ✓ How to create a storage account?
    ✓ How to create
        i. Azure BLOB storage
        ii. Azure file share
4. Azure Virtual Machine
    ✓ Create a windows virtual machine
    ✓ Create a linux virtual machine

Types of access on subscription:

- Owner
    - If any user gets the owner role, he/she is allowed to create, modify or delete anything within Azure.
- Contributor
    - If any user gets the contributor role, he/she is allowed to create, modify (but no delete) anything within Azure.
- Reader
    - If any user gets the reader role, he/she is allowed only to read (no create, no modify, no delete) anything within Azure.